Commitment and present bias: Experimental evidence

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Example Seminar Series
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Outline

Introduction

Model
- Setup
- Predictions

Design

Results

Conclusion

Introduction

Motivation

- People plan to act patiently, then take the immediate reward when the moment arrives (Doe 2020)

Motivation

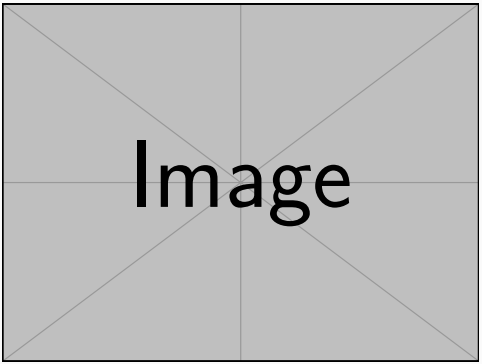
- People plan to act patiently, then take the immediate reward when the moment arrives (Doe 2020)
- **Present bias:** an extra discount on everything that is not “now” (Lee, Kim, and Park 2021)

Motivation

- People plan to act patiently, then take the immediate reward when the moment arrives (Doe 2020)
- **Present bias:** an extra discount on everything that is not “now” (Lee, Kim, and Park 2021)
- People who know they are present-biased seek commitment

This paper

- Laboratory experiment: intertemporal allocation decisions under one of two conditions
- Estimate a quasi-hyperbolic discounting model
- **Finding:** the treatment lowers present bias and raises demand for commitment



Model

Quasi-hyperbolic discounting

Present bias in one parameter

A decision maker values a stream of consumption as

$$U_t = u(x_t) + \beta \sum_{\tau=1}^{T} \delta^\tau u(x_{t+\tau})$$

- $\beta = 1$: exponential discounting
- $\beta < 1$: present bias

Example

With $\beta = 0.7$ and $\delta = 1$, \$110 tomorrow is worth \$77 today, so \$100 today wins. A week earlier, the same person prefers the \$110.

The value of commitment

Theorem

A present-biased person who knows that $\beta < 1$ strictly prefers to remove a tempting early option in advance.

Proof sketch.

Ahead of time, the person ranks the options using δ alone; at the moment of choice, the extra factor β reverses that ranking. Removing the early option prevents the reversal. □

Hypotheses

Hypothesis 1

The treatment lowers present bias: $\beta_{\text{treat}} < \beta_{\text{control}}$

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Hypothesis 2

The treatment raises take-up of the commitment device

Design

Experimental design

Control	Rewards shown as calendar dates
Treatment	Rewards shown as delays from today
Commitment	An option to lock in the later reward in advance

897 participants, randomly assigned to the two conditions.[1]

[1] Pre-registered at the AEA RCT Registry (AEARCTR-0000000).

Estimation

1. Estimate β and δ for each participant by maximum likelihood
2. Compare β across conditions, with standard errors that account for the two-stage estimator

Results

Main results

	Model 1	Model 2
Treatment effect	0.281	0.895
	(0.044)	(0.052)
Observations	897	897

Standard errors in parentheses.

▸ Robustness

Main results

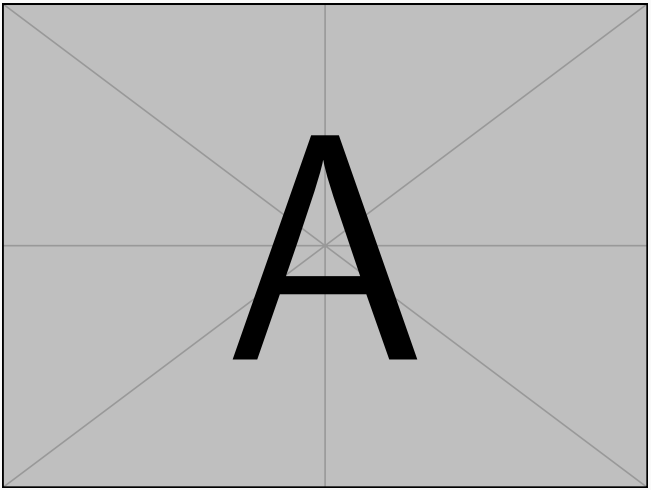
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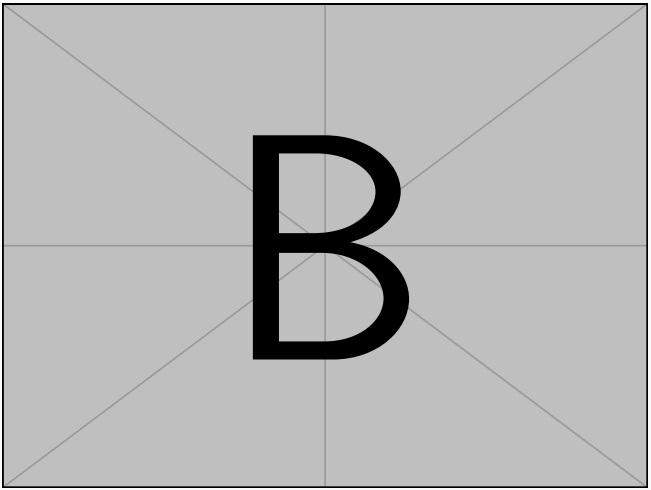
Model 1: an effect more than six standard errors from zero

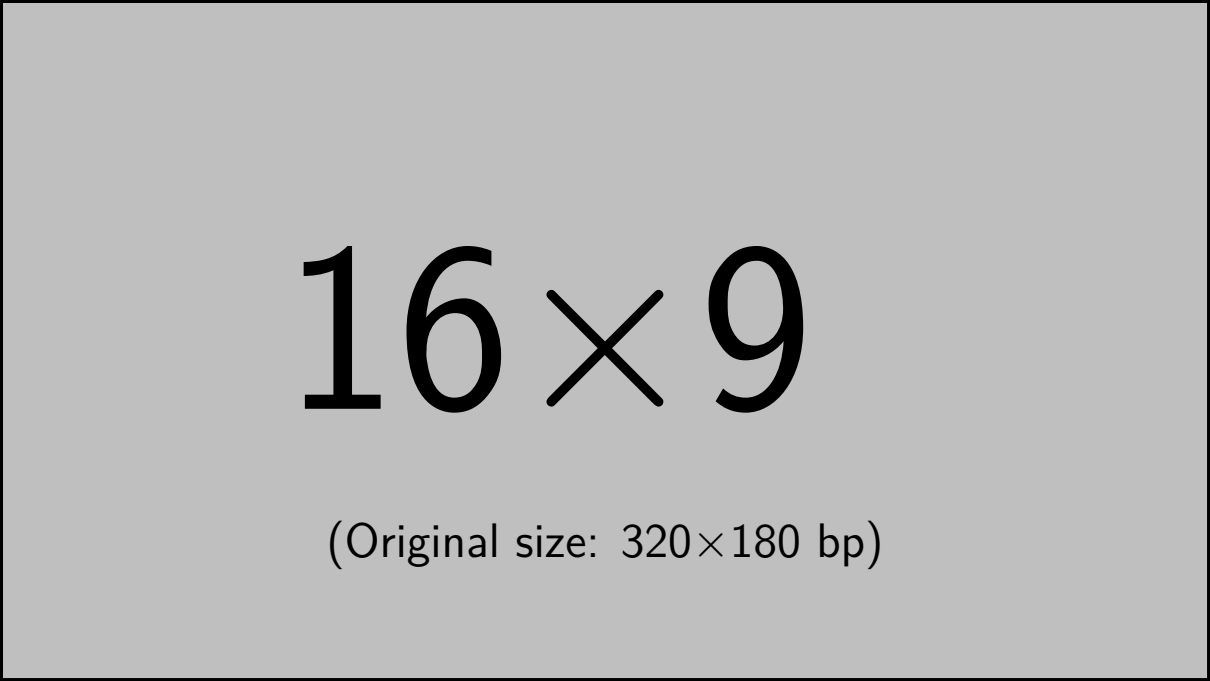
▸ Robustness

Commitment demand



Commitment demand




16×9
(Original size: 320×180 bp)

Conclusion

Takeaways

- A simple change in how future rewards are presented lowers present bias
- It also raises demand for commitment
  - Take-up of the commitment device rises in the treatment group

Caveat

A single laboratory sample; field replication is the natural next step.

Thank you

Paper: example.com/paper.pdf
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Robustness: excluding inattentive participants

	Model 1	Model 2
Treatment effect	0.264	0.871
	(0.049)	(0.058)
Observations	812	812

Standard errors in parentheses.

◂ Main results

Replication

Every table and figure is rebuilt from the project folder:

```
cd programs
Rscript main.R    # writes ../tables/ and ../figures/
```